

CERTAIN TREE SPECIES HAVE A SYMBIOSIS WITH FUNGI

THE UBIQUITY OF MYCORRHIZAE

IN CONTEXT

KEY FIGURE

Albert Frank (1839–1900)

BEFORE

1840 German botanist Theodor Hartig discovers a network of filaments on the roots of pine trees.

1874 Hellmuth Bruchmann, a German biologist, notes the “Hartig net” is made of fungal filaments.

AFTER

1937 A.B. Hatch, an American botanist, shows a beneficial relationship between pine trees and mycorrhizal fungus.

1950 Swedish botanists Elias Melin and Harald Nilsson show that plant roots can extract more nutrients from the soil with the aid of mycorrhizae.

1960 Another Swedish botanist, Erik Björkman, shows that plants pass carbon into mycorrhizal fungi in exchange for phosphate and nitrate.

In 1885, a professor of plant pathology at the Royal College of Agriculture in Berlin named Albert Frank was the first to see a connection between fungi growing on tree roots and the health of the trees. Frank realized that